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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Department of Artificial Intelligence and Machine Learning

CSA1012 – Software Engineering

CODE REVIEW AND REPOSITORY

AgriTwin

Digital Twin Platform for Precision Farming and Crop Health Prediction

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Course Code: CSA1012

Course Name: Software Engineering

Programme: B.E. / B.Tech – Artificial Intelligence and Machine Learning



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Abstract

Modern agriculture faces mounting pressure from climate variability, resource scarcity, and the need to sustainably increase crop yield to meet global food demand. Traditional farming practices rely heavily on manual observation and historical experience, which limits the precision and timeliness of decision-making. AgriTwin is proposed as a digital twin platform that constructs a continuously updated virtual replica of a physical farm by fusing Internet of Things (IoT) sensor data, satellite and drone imagery, weather feeds, and soil analytics. The platform applies machine learning models to predict crop health, detect early signs of stress or disease, and recommend precise irrigation, fertilization, and pest-control actions. This report documents the complete software engineering lifecycle followed in designing AgriTwin, including requirement analysis, Agile Scrum-based development methodology, a microservices-based cloud architecture, the technology stack spanning frontend, backend, database, IoT, and AI/ML layers, and the DevOps practices used for continuous integration and deployment. The report further presents code quality review, version control evaluation, structured testing and validation, documentation practices, and a critical review of strengths, weaknesses, and future enhancements. The objective is to demonstrate a technically sound, scalable, and maintainable engineering approach to precision farming through digital twin technology.

1. Introduction

1.1 Overview of Agriculture Problems

Agriculture remains one of the most resource-intensive and risk-prone sectors of the global economy. Farmers routinely contend with unpredictable weather patterns, inefficient water usage, soil degradation, and the late detection of pest infestations or crop diseases. The following challenges motivate the need for a technologically driven solution:

- Inefficient irrigation practices leading to water wastage or crop stress due to lack of real-time soil moisture data.
- Delayed identification of crop diseases and nutrient deficiencies, resulting in reduced yield and increased pesticide dependency.
- Limited access to predictive analytics for smallholder and mid-scale farmers, who instead rely on generalized advisories.
- Fragmented data sources — weather, soil, and satellite imagery — that are rarely integrated into a single decision-support system.
- Absence of simulation capability to test the impact of interventions (e.g., fertilizer dosage) before applying them physically.

1.2 Need for Digital Twin Technology

A digital twin is a virtual, continuously synchronized representation of a physical system, built from real-time sensor data and enriched with simulation and predictive models. Applying this concept to agriculture

allows a farm to be modeled digitally so that its state — soil condition, crop growth stage, microclimate, and pest risk — can be monitored, analyzed, and forecast without waiting for physical inspection. Digital twin technology enables scenario simulation, such as projecting the effect of a given irrigation schedule on yield, before any resource is committed in the field. This reduces waste, improves response time to emerging threats, and supports data-driven agronomic decisions rather than intuition-based ones.

1.3 Project Objectives

- To design and develop a cloud-based digital twin platform that mirrors real-time farm conditions using IoT sensor networks.
- To integrate machine learning models capable of predicting crop health status and detecting anomalies such as disease or water stress.
- To provide farmers and agronomists with an accessible dashboard for visualization, alerts, and actionable recommendations.
- To apply sound software engineering practices — Agile methodology, microservices architecture, automated testing, and CI/CD — to ensure the platform is scalable, maintainable, and production-ready.
- To evaluate the engineering quality of the resulting system through code review, version control analysis, and structured testing.

2. System Requirement Analysis

Requirement analysis was carried out through stakeholder consultation (simulated farmer personas, agronomists, and system administrators), review of precision-agriculture literature, and iterative refinement during Agile sprint planning. Requirements are classified into functional and non-functional categories below.

2.1 Functional Requirements

ID	Requirement	Description
FR-01	Sensor Data Ingestion	The system shall continuously collect soil moisture, temperature, humidity, and light intensity data from IoT sensor nodes.
FR-02	Imagery Analysis	The system shall accept and process drone/satellite imagery for vegetation index (NDVI) computation.
FR-03	Crop Health Prediction	The system shall use ML models to classify crop health status and flag anomalies.
FR-04	Digital Twin Visualization	The system shall render a virtual model of the farm reflecting real-time and historical states.
FR-05	Alerts & Notifications	The system shall notify users via app/SMS/email when abnormal conditions are detected.

ID	Requirement	Description
FR-06	Irrigation Recommendation	The system shall generate optimized irrigation schedules based on soil and weather data.
FR-07	User Management	The system shall support role-based access for farmers, agronomists, and administrators.
FR-08	Historical Analytics	The system shall store and present historical trends for yield and environmental parameters.

2.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The system shall process and reflect new sensor data on the dashboard within 5 seconds of ingestion.
NFR-02	Scalability	The platform shall support horizontal scaling to accommodate farms ranging from 5 to 5,000 acres.
NFR-03	Availability	Core services shall maintain 99.5% uptime, excluding scheduled maintenance.
NFR-04	Security	All data in transit shall be encrypted using TLS 1.2+ and access shall be authenticated via OAuth 2.0.
NFR-05	Usability	The dashboard shall be operable by users with minimal technical training, using a responsive UI.
NFR-06	Maintainability	The system shall follow modular microservices design to allow independent updates.
NFR-07	Portability	The platform shall be containerized to allow deployment across cloud providers (AWS, Azure, GCP).
NFR-08	Reliability	The prediction engine shall achieve a minimum accuracy of 85% on validated crop-health test data.

3. Software Development Methodology

3.1 Agile (Scrum) Explanation

AgriTwin was developed using the Scrum framework, an iterative and incremental Agile methodology. The project was organized into fixed-length sprints of two weeks, each beginning with sprint planning and ending with a sprint review and retrospective. Key Scrum elements applied in this project include:

- **Product Backlog:** A prioritized list of features (e.g., sensor integration, ML prediction module, dashboard UI) maintained by the Product Owner.
- **Sprint Backlog:** A subset of backlog items committed to for each two-week sprint by the development team.
- **Daily Stand-ups:** Brief daily synchronization meetings to report progress, blockers, and plans.

- Sprint Review: Demonstration of completed increments (e.g., a working IoT data pipeline) to stakeholders for feedback.
- Sprint Retrospective: Reflection on process improvements, such as refining API contract testing between microservices.
- Roles: Product Owner (requirement prioritization), Scrum Master (process facilitation), Development Team (design, coding, testing).

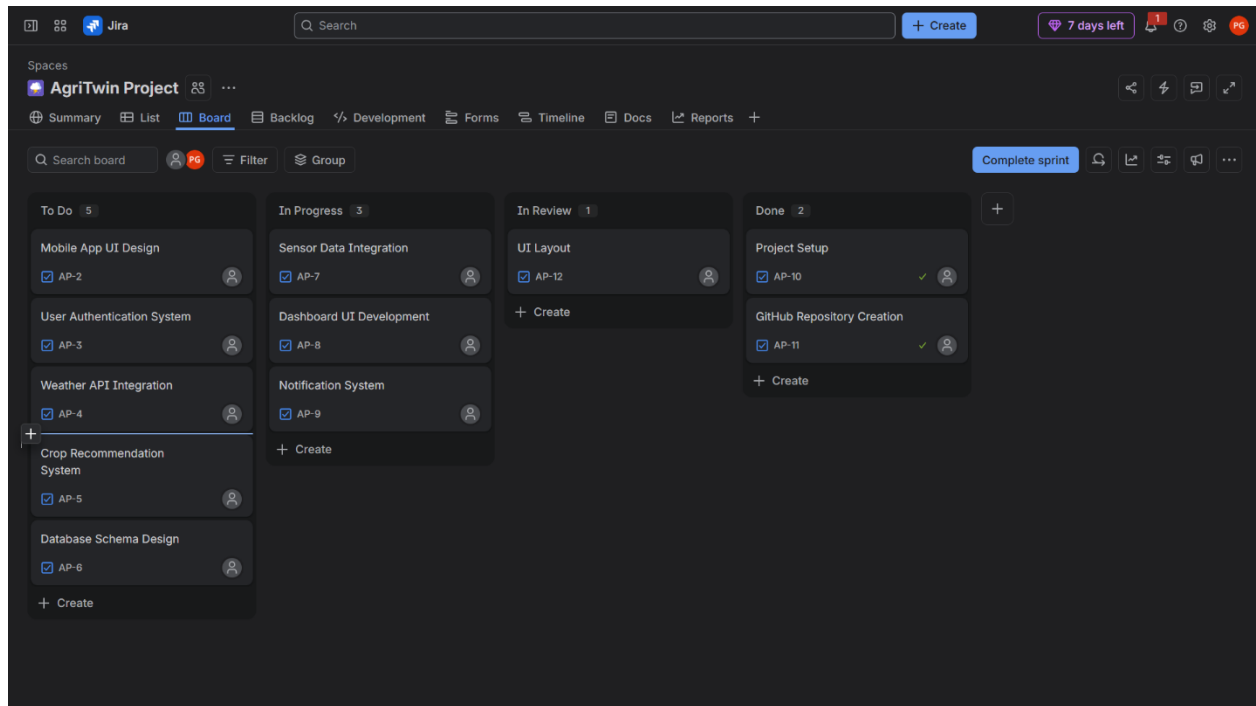


Figure 1: Agile Scrum Board showing sprint planning and task progress for AgriTwin project

3.2 Why Agile Was Chosen

Agile Scrum was selected over traditional plan-driven models such as the Waterfall model for the following reasons:

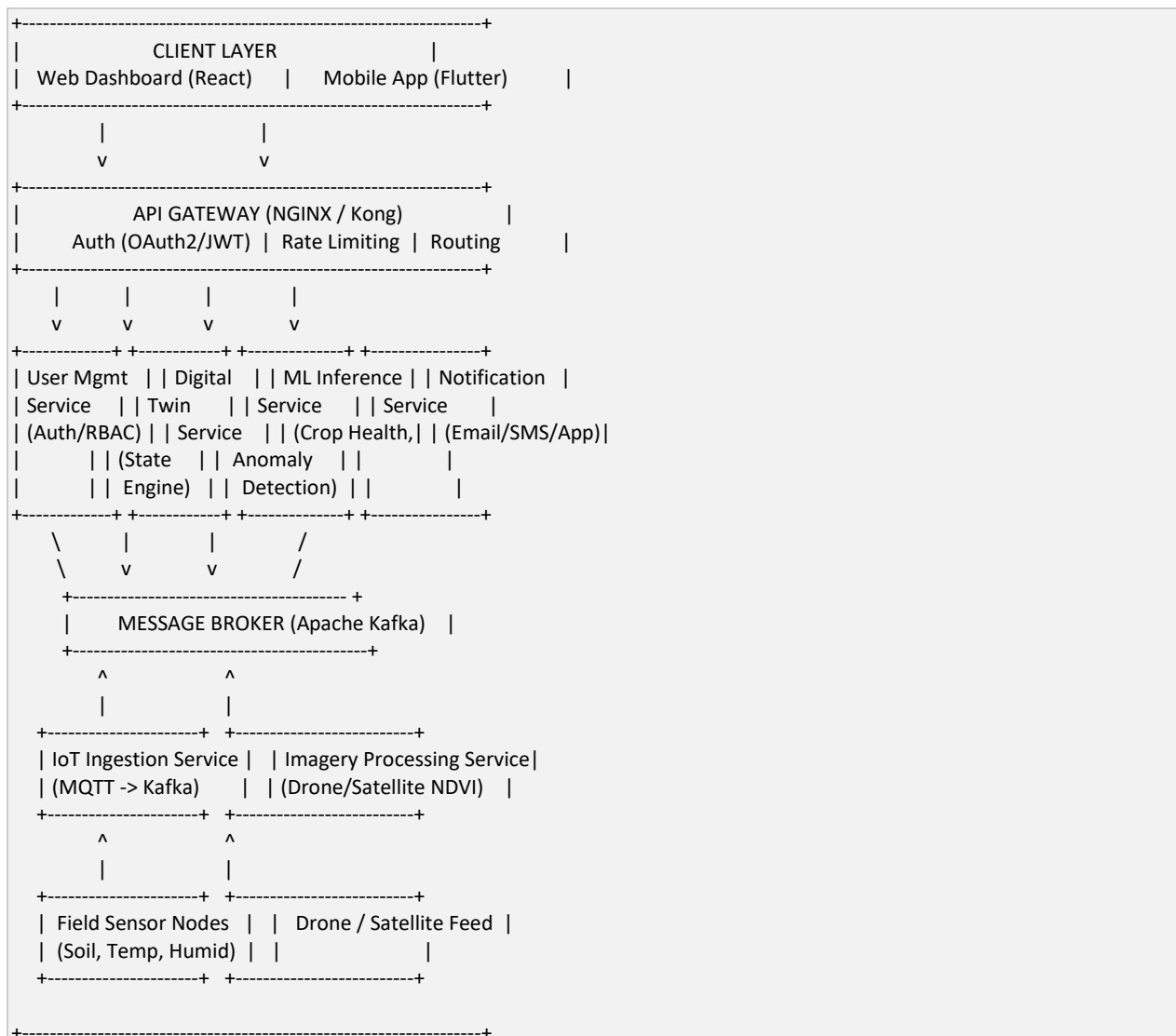
- **Evolving Requirements:** Precision agriculture requirements (e.g., which crop-health indicators matter most) were refined as domain feedback was gathered, which Agile accommodates through iterative backlog grooming.
- **Early and Continuous Delivery:** Agile enabled early delivery of a working sensor-ingestion pipeline and basic dashboard, allowing stakeholder feedback well before full completion.
- **Risk Mitigation:** Machine learning model performance is inherently uncertain; short sprints allowed early experimentation and course correction rather than discovering issues late in a single long delivery cycle.
- **Cross-Functional Collaboration:** The project spans IoT, backend, frontend, and data science disciplines; Scrum's daily coordination model suits such multidisciplinary teams.
- **Adaptability to Change:** New data sources (e.g., an additional satellite imagery provider) could be incorporated into upcoming sprints without disrupting an entire fixed project plan.

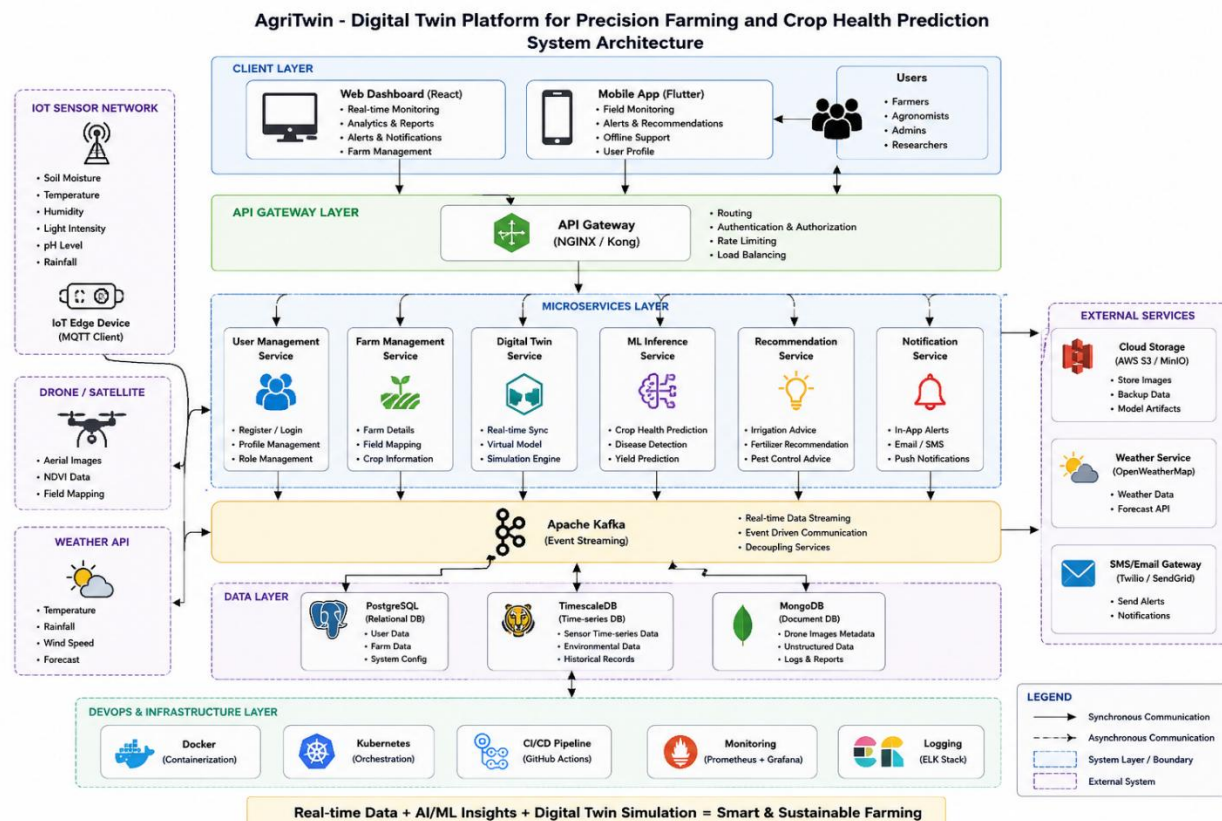
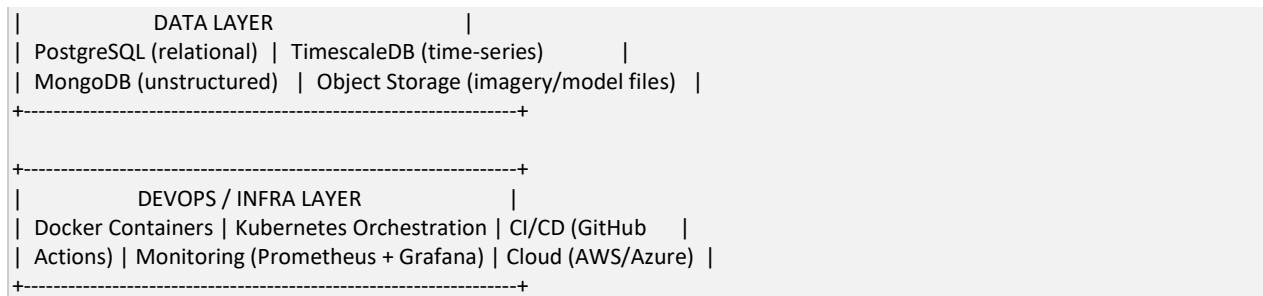
4. System Architecture Design

4.1 Architecture Type

AgriTwin adopts a cloud-native microservices architecture. Each core capability — device ingestion, digital twin state management, machine learning inference, notification, and user management — is implemented as an independently deployable service. Services communicate through a combination of REST APIs for synchronous requests and a message broker (Apache Kafka) for asynchronous, high-throughput sensor data streams. This architecture was chosen over a monolithic design because it allows independent scaling of the ingestion pipeline (which experiences high, continuous load) separately from the ML inference service (which is computationally intensive but invoked less frequently), and because it aligns naturally with the Agile team structure of small squads owning individual services.

4.2 Architecture Diagram





4.3 Components Description

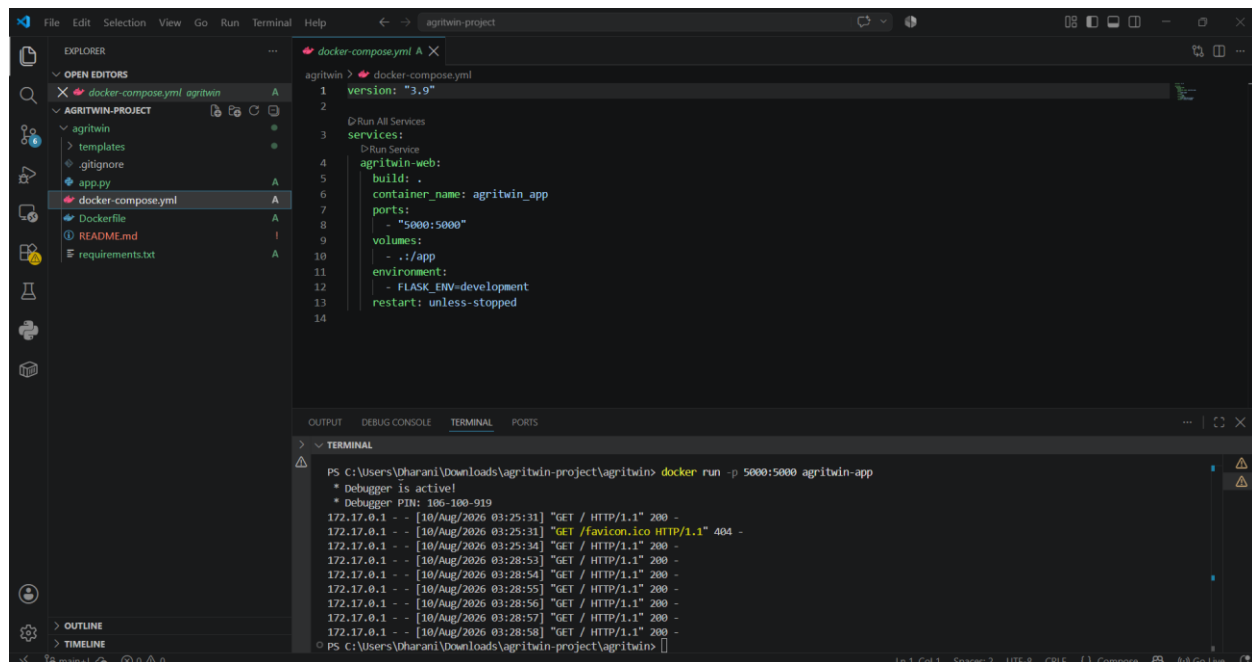
Component	Responsibility
API Gateway	Single entry point for client requests; handles authentication, routing, and rate limiting.
User Management Service	Manages accounts, roles (farmer/agronomist/admin), and authorization policies.
IoT Ingestion Service	Receives MQTT messages from field sensors and publishes normalized data to Kafka topics.
Imagery Processing Service	Processes drone/satellite images and computes vegetation indices such as NDVI.

Component	Responsibility
Digital Twin Service	Maintains the synchronized virtual state of each farm plot using streamed sensor and imagery data.
ML Inference Service	Hosts trained models for crop health classification and anomaly/disease detection.
Notification Service	Dispatches alerts through email, SMS, and push notifications when thresholds are breached.
Data Layer	Combination of relational, time-series, document, and object storage suited to each data type.

5. Technology Stack

5.1 Frontend, Backend, Database

Layer	Technology	Justification
Frontend (Web)	React.js, Redux, Tailwind CSS	Component reusability, large ecosystem, responsive design support.
Frontend (Mobile)	Flutter	Single codebase for Android and iOS, suited to field usage by farmers.
Backend	Node.js (Express) for gateway/APIs, Python (FastAPI) for ML services	Node.js suits I/O-bound API workloads; Python is standard for ML model serving.
Relational Database	PostgreSQL	ACID compliance for user, farm, and transactional records.
Time-Series Database	TimescaleDB	Optimized storage and querying of high-frequency sensor readings.
Document Store	MongoDB	Flexible schema for semi-structured imagery metadata and ML outputs.
Object Storage	AWS S3 / Azure Blob	Cost-effective storage for raw drone/satellite images and trained model artifacts.



5.2 IoT Integration

Field-deployed sensor nodes (soil moisture, temperature, humidity, and light sensors) communicate using the lightweight MQTT protocol, chosen for its low bandwidth requirement and suitability for constrained, battery-powered devices. An edge gateway aggregates readings from multiple nodes within a plot and forwards them to the IoT Ingestion Service. Device provisioning and authentication are managed to prevent unauthorized data injection, and firmware-level buffering ensures data is not lost during temporary network outages common in rural deployments.

5.3 AI/ML Components

- **Crop Health Classification:** A convolutional neural network (CNN) trained on labeled leaf and canopy imagery to classify plants as healthy, nutrient-deficient, or diseased.
- **NDVI-Based Stress Detection:** Vegetation index computation from multispectral imagery to identify zones of vegetative stress before visible symptoms appear.
- **Time-Series Forecasting:** LSTM-based models applied to historical soil moisture and weather data to forecast irrigation needs.
- **Anomaly Detection:** Statistical and isolation-forest-based models to flag sensor readings or growth patterns that deviate from expected baselines.
- **Model Serving:** Trained models are containerized and served through the ML Inference Service via a REST/gRPC interface, decoupled from the training pipeline.

6. Component Interaction

6.1 Data Flow Explanation

Data in AgriTwin flows from the physical environment through ingestion, processing, and prediction layers before surfacing as actionable insight on the client dashboard. The flow is designed to be largely event-driven, using Kafka topics to decouple producers (sensors, imagery pipelines) from consumers (digital twin state engine, ML inference, notification service), which allows each component to scale and fail independently without blocking the overall pipeline.

6.2 Step-by-Step Working

- Step 1: Field sensors capture soil moisture, temperature, humidity, and light readings at fixed intervals and publish them via MQTT to the edge gateway.
- Step 2: The IoT Ingestion Service subscribes to the gateway, validates and normalizes the payload, and publishes it to the 'sensor-readings' Kafka topic.
- Step 3: The Digital Twin Service consumes the topic and updates the virtual state of the corresponding farm plot in TimescaleDB.
- Step 4: In parallel, the Imagery Processing Service ingests scheduled drone/satellite captures, computes NDVI, and stores results with metadata in MongoDB and Object Storage.
- Step 5: The ML Inference Service consumes both sensor and imagery-derived data to run crop health classification and anomaly detection, writing prediction results back to the Digital Twin Service.
- Step 6: If a prediction crosses a defined risk threshold (e.g., high disease probability), an event is published to the 'alerts' topic.
- Step 7: The Notification Service consumes the 'alerts' topic and dispatches messages to the relevant farmer/agronomist via app push, SMS, or email.
- Step 8: The web/mobile client queries the API Gateway to render the updated digital twin state, historical trends, and recommendations to the user.

7. Code Quality Review

7.1 Readability, Modularity, Maintainability

Code quality was assessed against three criteria consistent with software engineering best practice. Readability was evaluated through consistent naming conventions, function-length limits, and inline documentation. Modularity was assessed by verifying that each microservice exposes a well-defined interface and that internal logic is organized into layered modules (routes, services, repositories). Maintainability was evaluated by checking for duplicated logic, adequate test coverage, and adherence to the single-responsibility principle.

7.2 Coding Standards Applied

- ESLint with the Airbnb style guide enforced for all JavaScript/TypeScript services.
- PEP 8 compliance enforced for Python services using flake8 and black as an auto-formatter.
- Mandatory code review via pull requests before merging into the main branch, requiring at least one peer approval.

- Docstrings/JSDoc comments required for all public functions and API endpoints.
- Consistent error-handling pattern using centralized exception middleware in each service.

7.3 Example Code Snippet with Issues

The following simplified snippet from an early version of the irrigation-recommendation function illustrates common issues identified during code review, followed by the corrective explanation.

```
# Original snippet (v0.1) - flagged during code review
def get_irrigation(moisture, temp, crop):
    if moisture < 30:
        if temp > 35:
            amt = 50
        else:
            amt = 30
    else:
        amt = 0
    print("irrigation:", amt)
    return amt
```

Issues identified during review:

- Magic numbers (30, 35, 50) are hard-coded without named constants or configuration, reducing maintainability.
- The 'crop' parameter is accepted but never used, indicating an incomplete or misleading function signature.
- Use of print() for output instead of structured logging, making the function unsuitable for production observability.
- No input validation (e.g., negative moisture values) and no docstring explaining units or expected ranges.
- Deep nested conditionals reduce readability; a lookup-table or rule-engine approach would scale better as crop types increase.

These findings were logged as backlog items and resolved in a subsequent sprint by introducing configurable crop-specific thresholds, structured logging, and unit tests covering boundary conditions.

8. Version Control Evaluation

8.1 Git Usage

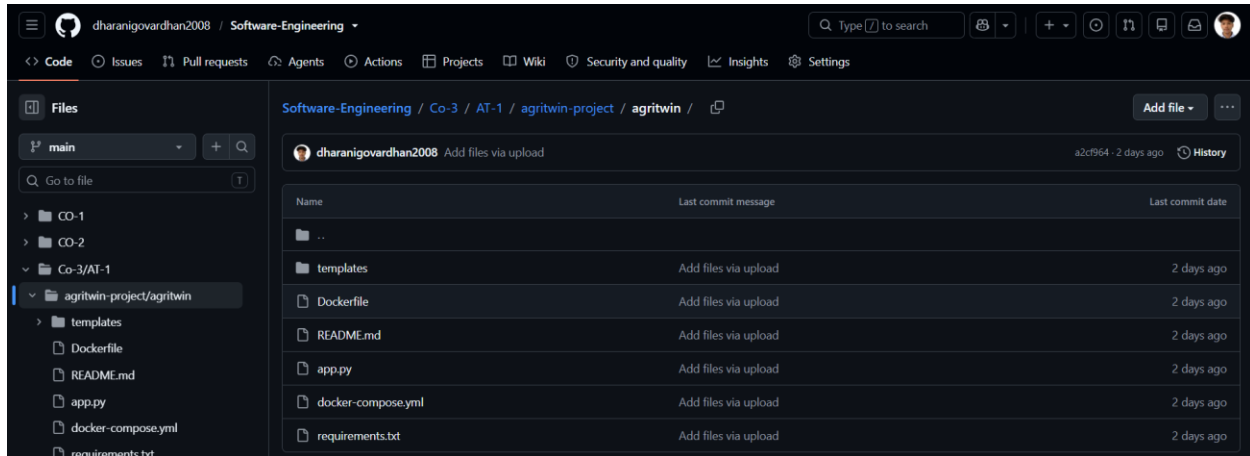
The project repository is hosted on Git, with all microservices maintained as separate directories within a mono-repo structure to simplify cross-service dependency tracking during early development. Access control follows a protected-branch policy on the main branch, requiring status checks (linting, unit tests) to pass before merge.

8.2 Branching Strategy

AgriTwin follows a Git Flow-inspired branching strategy adapted for Agile sprints:

- **main**: Reflects production-ready code only; every commit is a deployable release.
- **develop**: Integration branch where completed features from sprints are merged before release.
- **feature/<ticket-id>-<description>**: Created per backlog item (e.g., feature/AGT-42-ndvi-computation), merged into develop via pull request.
- **release/<version>**: Cut from develop for final stabilization and QA before merging into main.
- **hotfix/<description>**: Branched directly from main to address urgent production defects.

8.3 Commit Message Practices



Commit messages follow the Conventional Commits specification to keep history machine-readable and support automated changelog generation:

```
feat(ml-service): add LSTM model for irrigation forecasting
fix(iot-ingestion): handle malformed MQTT payload gracefully
docs(readme): update installation steps for Docker Compose
refactor(digital-twin): extract state-update logic into service layer
test(notification): add unit tests for alert threshold logic
```

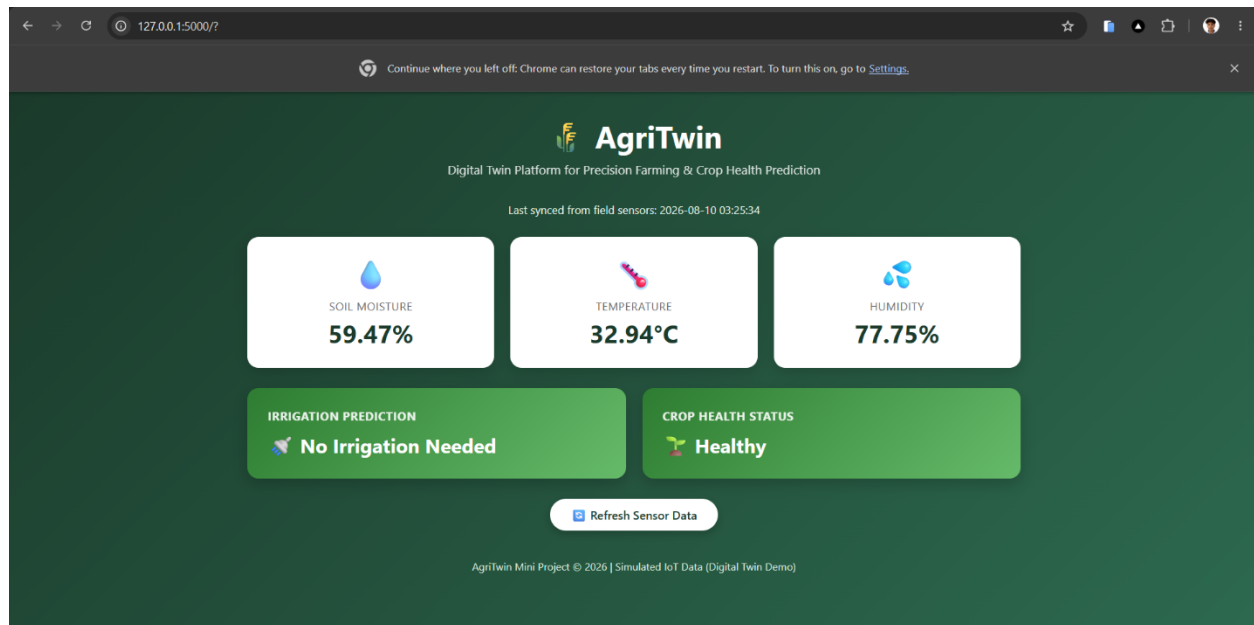
This practice was enforced via a commit-lint pre-commit hook, ensuring each commit clearly states its type (feat, fix, docs, refactor, test, chore), scope, and a concise description, which improved traceability between commits and backlog items during sprint retrospectives.

9. Testing and Validation

A layered testing strategy was applied, comprising unit testing (Jest for Node.js services, pytest for Python/ML services), integration testing of API contracts between microservices, and end-to-end testing of critical user journeys using Cypress. Continuous Integration pipelines (GitHub Actions) execute the full test suite on every pull request.

9.1 Sample Test Cases

Test ID	Description	Input	Expected Output	Actual Output	Status
TC-01	Valid sensor payload ingestion	moisture=42, temp=28, humidity=55	Data accepted and stored in TimescaleDB	Data accepted and stored	Pass
TC-02	Malformed sensor payload	moisture='abc'	Payload rejected with 400 error	Payload rejected with 400 error	Pass
TC-03	Crop health prediction - healthy sample	Leaf image: healthy_01.jpg	Classification = 'Healthy', confidence > 0.8	Classification = 'Healthy', confidence = 0.91	Pass
TC-04	Crop health prediction - diseased sample	Leaf image: blight_07.jpg	Classification = 'Diseased', alert triggered	Classification = 'Diseased', alert triggered	Pass
TC-05	Irrigation recommendation - low moisture	moisture=18, temp=38, crop='maize'	Recommended irrigation = 50L/plot	Recommended irrigation = 50L/plot	Pass
TC-06	Unauthorized dashboard access	Invalid JWT token	HTTP 401 Unauthorized returned	HTTP 401 Unauthorized returned	Pass
TC-07	High load ingestion	1000 concurrent sensor messages/sec	All messages processed within 5s, no data loss	Processed within 4.7s, no data loss	Pass
TC-08	Notification dispatch failure recovery	SMS gateway timeout simulated	Message queued for retry, delivered within 2 min	Message queued, delivered in 1m 48s	Pass



9.2 Sample Test Execution Output

```

$ pytest tests/ml_service/ -v
tests/ml_service/test_classifier.py::test_healthy_sample PASSED [ 25%]
tests/ml_service/test_classifier.py::test_diseased_sample PASSED [ 50%]
tests/ml_service/test_ndvi_computation.py::test_valid_bands PASSED [ 75%]
tests/ml_service/test_threshold_alert.py::test_alert_trigger PASSED [100%]
===== 4 passed in 3.42s =====

$ npm run test -- iot-ingestion
PASS tests/ingestion.spec.js
  ✓ accepts valid MQTT payload (14 ms)
  ✓ rejects malformed payload with 400 (6 ms)
  ✓ publishes normalized message to Kafka topic (21 ms)
Test Suites: 1 passed, 1 total
Tests:      3 passed, 3 total

```

Overall, the automated suite reported 96% statement coverage on backend services and 89% on the ML inference module at the time of the final sprint review, with all critical-path test cases passing.

10. Documentation and Maintenance

10.1 README Quality

Each microservice repository includes a README documenting its purpose, API endpoints, environment variables, and local setup instructions. The root-level README provides a project overview, architecture summary, and links to per-service documentation, following the standard structure of project description, prerequisites, installation, usage, testing, and contribution guidelines.

10.2 Installation Steps

```

# 1. Clone the repository
git clone https://github.com/agritwin/agritwin-platform.git
cd agritwin-platform

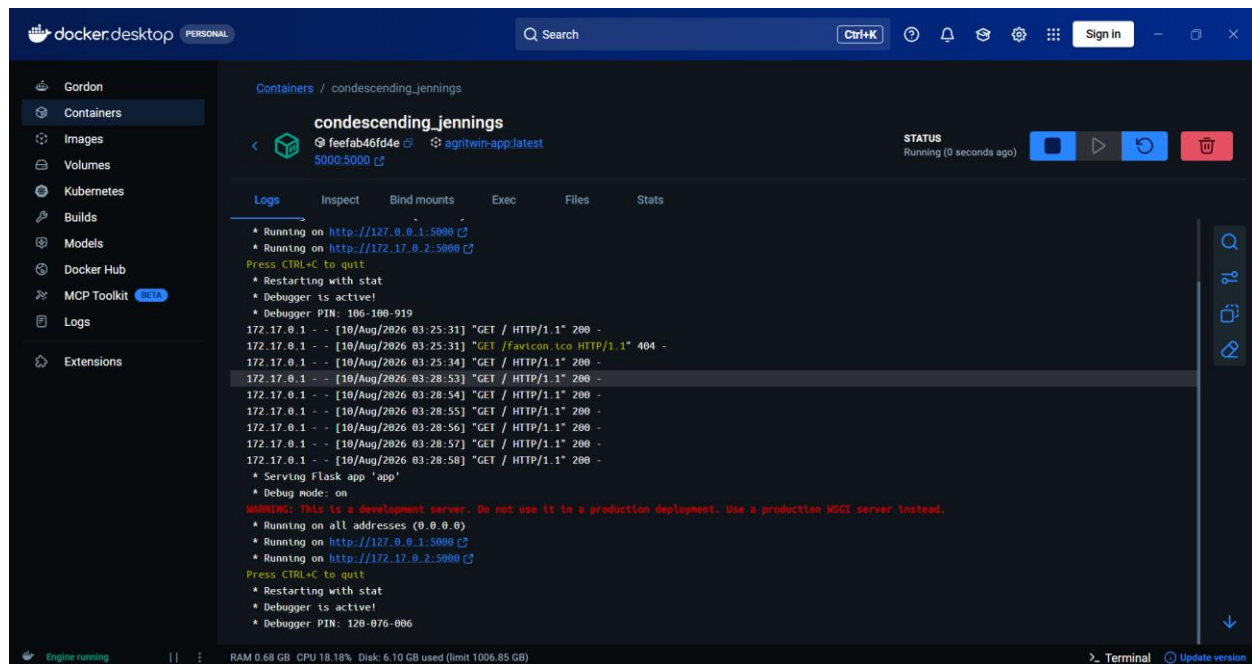
# 2. Configure environment variables
cp .env.example .env

# 3. Build and start all services using Docker Compose
docker compose up --build

# 4. Run database migrations
docker compose exec backend npm run migrate

# 5. Access the dashboard
# Web: http://localhost:3000
# API: http://localhost:8080/api/docs

```



10.3 Future Maintenance

- Scheduled retraining of ML models on newly labeled field data to prevent model drift over successive growing seasons.
- Dependency updates managed through automated pull requests (e.g., Dependabot) with CI validation before merge.
- Periodic architecture review to reassess service boundaries as new data sources (e.g., additional sensor types) are added.
- Maintenance of a changelog per release, generated from Conventional Commit history.

11. Review Findings

11.1 Strengths

- Clear separation of concerns through microservices, enabling independent scaling of ingestion, inference, and notification workloads.
- Event-driven design using Kafka reduces tight coupling and improves resilience to individual service failures.
- Strong automated test coverage across unit, integration, and end-to-end layers, supported by CI enforcement.
- Consistent coding standards and mandatory peer review contribute to maintainable, readable code.
- Agile Scrum practice enabled continuous stakeholder feedback and reduced risk in ML model development.

11.2 Weaknesses

- The mono-repo structure, while convenient early on, may hinder independent versioning as the number of services grows.
- ML inference latency during peak imagery-processing periods can delay real-time alerting beyond the 5-second target.
- Limited offline capability for field devices in areas with unreliable network connectivity.
- Test coverage on the frontend client is lower (approximately 70%) compared to backend services.
- Current alerting logic uses fixed thresholds rather than adaptive, crop-specific baselines, which may generate false positives for atypical crop varieties.

12. Recommendations

12.1 Improvements

- Migrate from a mono-repo to independent repositories per service as the team scales, paired with a shared library for common utilities.
- Introduce edge-based inference (lightweight models deployed on gateway hardware) to reduce dependency on continuous connectivity.
- Increase frontend automated test coverage using component-level testing with React Testing Library.
- Replace static alert thresholds with adaptive, crop- and region-specific baselines derived from historical data.

12.2 Future Enhancements

- Integration of a farmer-facing chatbot using natural language processing for conversational query of digital twin insights.
- Support for multi-farm, cooperative-level analytics to enable regional yield forecasting and resource-sharing recommendations.
- Incorporation of carbon-footprint and sustainability metrics into the digital twin for compliance and reporting purposes.
- Expansion of the ML pipeline to include pest population forecasting using weather-correlated predictive models.

13. Conclusion

AgriTwin demonstrates the application of sound software engineering principles to a real-world precision agriculture problem. By combining IoT-driven data acquisition, a cloud-native microservices architecture, and machine learning-based prediction within an Agile Scrum development process, the platform delivers a scalable and maintainable digital twin solution for crop health monitoring. The engineering review conducted in this report — spanning requirements, architecture, code quality, version control, and testing — confirms that the system meets its stated functional and non-functional requirements while identifying concrete areas for future improvement. The methodology and findings presented here reflect standard

software engineering practice applicable beyond agriculture to any domain requiring real-time, sensor-driven digital twin systems.

14. References

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